1. Project Overview

The system is a **web-based product management and verification platform** with **role-based access control**. It will manage product specifications, approvals, and real-world product verification using **QR code scanning**.

The system will allow:

- Product data entry through **manual entry or Excel import**
- **Multi-level verification and approval**
- **Role-based user management**
- **QR code-based product verification during production or inspection**
- **Document and field level access control**

The objective is to **ensure product specifications match the approved design specifications** before production or delivery.

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2. User Roles & Permissions

The system will have **five user roles**:

1. Super Admin

Highest level access.

Responsibilities:

- Full system control
- Create and manage admins
- Configure system settings
- Manage role permissions

Permissions:

- Full access to users
- Full access to product records
- Full access to documents
- System configuration

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2. Admin

Administrative user with operational control.

Responsibilities:

- Create and manage users

- Assign roles
- Control user permissions
- Manage product master data

Permissions:

- Create/Edit/Delete users
- Control which fields users can view
- Control which documents users can upload/view
- Manage product records

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3. Production User

User responsible for handling **physical products on the production line**.

Responsibilities:

- Scan product QR codes
- Enter actual product specifications
- Compare scanned product with approved product specifications

Permissions:

- Scan QR code
- Fill product parameters
- System validation against approved records
- View verification results

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4. Quality User

Responsible for **verification and approval of product specifications**.

Responsibilities:

- Verify product records entered in the system
- Approve product data

Permissions:

- View pending records
- Verify product data
- Approve or reject product specifications

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5. Viewer

Read-only access.

Permissions:

- View product records
- View verification results
- Cannot edit data

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3. System Modules

3.1 User Management Module

Admin can:

- Create new users
- Assign roles
- Set login credentials
- Control user permissions

Admin can also configure:

- Which **fields** users can view
- Which **documents** users can upload/view

Example fields:

- Tube Length
- Tube Diameter
- Part Type
- PDC Length
- Flange Yoke
- Coupling Flange
- Balancing RPM

Example documents:

- Drawing Document
- IQA Document
- PO Document
- TRSO Document

3.2 Product Data Management

Product specifications will be stored in a **Product Master Database**.

Product records can be added through:

Manual Entry

User fills form fields.

Excel Import

User uploads Excel file containing product specifications.

Example fields from Excel:

- Part Number
- Customer Name
- Tube Dia & Thickness
- Tube Length
- Part Type
- Noise Deadener
- Coupling Flange
- Balancing RPM
- Total Length
- Unbalance Value
- Vendor Code

When product data is entered:

Initial status = **Pending**

3.3 Product Verification Workflow

Product data goes through the following workflow:

Step 1 — Data Entry

User enters product data manually or via Excel.

Status = **Pending**

Step 2 — Verification

Quality user checks product data.

Possible actions:

- Verify record
- Reject record

Status = **Verified**

Step 3 — Approval

After verification, the record is approved and stored in the **Approved Product Master Table**.

Status = **Approved**

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4. QR Code Based Product Validation

Each physical product will have a **QR code**.

The QR code contains:

- Part Number
- Possibly additional identifiers

Production Workflow

1. Production user scans QR code.
2. System extracts **Part Number**.
3. System fetches **approved product specification** from database.
4. Production user enters **actual measured specifications**.

Example:

- Length
- Diameter
- Flange orientation
- Balancing values

Data Comparison

System compares:

Actual Product Data
vs
Approved Product Specification

Result

If values match →
Status = Verified

If values do not match →
Status = Not Verified / Error Notification

System displays notification accordingly.

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5. Dashboard Features

The system dashboard will show:

- Total products
- Pending approvals
- Verified records
- Product verification history
- Production verification logs

6. Key System Features

- Role-based access control
- Excel import of product data
- Multi-stage product approval workflow
- QR code scanning
- Real-time product verification
- Notification system
- Document management
- Audit logs